

LETTER TO THE EDITOR

Rational harm-benefit assessments by age group are required for continued COVID-19 vaccination

To the editor,

We read with interest the letter by Dr. Gül and Dr. Öztürk,¹ which comments about the previous letter by Dr. Polykretis.² The letter by Dr. Polykretis aimed to underline the differences between the genetic vaccines against COVID-19 and vaccines based on inactivated or attenuated viruses in terms of immunization mechanism. Moreover, and most importantly, it sought to emphasize the necessity of biodistribution studies in front of the numerous publications reporting on a variety of serious adverse events among vaccinees.² Considering that some pharmaceutical companies, such as Pfizer/BioNTech, had 'to move at the speed of science, to really understand what is taking place in the market' to release the vaccines (as declared later on by Janine Small, President of International Developed Markets, to the European Parliament on Monday, October 10th, 2022), there is nothing of scientifically despicable or misleading in seeking for the collection of more accurate data about biodistribution.

Dr. Gül and Dr. Öztürk accuse the letter by Dr. Polykretis of being 'misinforming' and of containing some 'basic errors', arguing on the definitions of genetic vaccines and autoimmunity. We would like to address on both cases. Regarding the definition of genetic vaccines, the letter by Dr. Polykretis is not misleading, as scientific literature reports that: 'gene vaccines are a new approach to immunization and immunotherapy in which, rather than a live or inactivated organism (or a subunit thereof), one or more genes that encode proteins of the pathogen are delivered'.³ As concerns the term autoimmunity, the Merriam-Webster medical dictionary it defines it as: 'a condition in which the body produces an immune response against its own tissue constituents'. Therefore, it is not misinforming or erroneous to define autoimmune reaction the response of the immune system against human cells that intake the lipid nanoparticles (LNPs) and translate the spike protein (in case of the mRNA vaccines), or that get infected by the adenovirus and express and translate the spike protein (in case of the adenovirus-based vaccines). Regarding the fact that even the 'traditional vaccines' cause the immune system to respond by attacking self-cells during

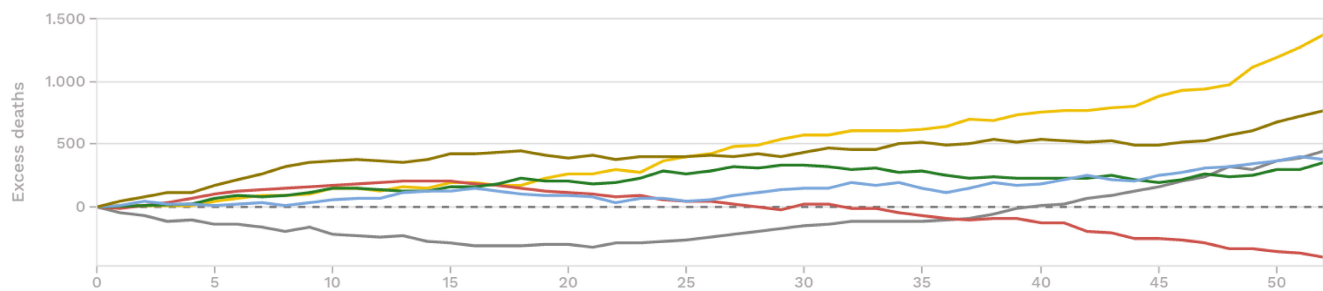
the immunization process, there are some fundamental aspects that should be underlined: (i) The vaccines based on inactivated or killed viruses involve principally presentation to antigen presenting cells (APCs) including macrophages, monocytes, B cells and dendritic cells that phagocytose the virus particles and present the viral antigens to CD4⁺ T-cells. The aforementioned classes of cells carry out this specific role within the organism, making them somewhat expendable, as there is a continuous turnover of such cells. (ii) The attenuated viruses have a reduced virulence and thus, the resulting infection involves a minor number of human cells. Instead, several sources of histopathological evidence demonstrate that the genetic vaccines exhibit an off-target distribution in tissues, which are terminally differentiated and subject to symptomatic injury. These include the heart and brain, which may sustain a massive production of spike protein which elicits a strong autoimmunological inflammatory response.^{4,5} The above mentioned histopathological findings confirm exactly the mechanism previously theorized by Dr. Polykretis: "For instance, if the mRNA contained in the LNPs would get internalized by cardiac myocytes, and such cells would produce the spike protein, the resulting inflammation would likely lead to the necrosis of the myocardium, with an extent proportional to the number of involved cells".²

An independent secondary analysis of serious adverse events reported in phase III clinical trials of Pfizer and Moderna, found that the mRNA vaccines combined were associated with an excess risk of serious adverse events of 1 per 800 vaccinated individuals.⁶ Nevertheless, indiscriminate COVID-19 vaccination has been expanded to include age groups and naturally immune with minimal chance of suffering major complications due to COVID-19. In these groups COVID-19 vaccination is not clinically indicated nor medically necessary. According to a large-scale risk-benefit analysis, between 31 207 and 42 836 young adults aged 18-29 years would need to receive a third mRNA vaccine dose to prevent one COVID-19 hospitalization over a course of six months.⁷ The authors estimate that at least 18.5 serious adverse events could occur for every

--- Baseline — 2017 — 2018 — 2019 — 2020 — 2021 — 2022

A

0-14 years



— Pooled deaths — Normal range --- Baseline - - - Substantial increase — Corrected for delay in registration

B

0-14 years

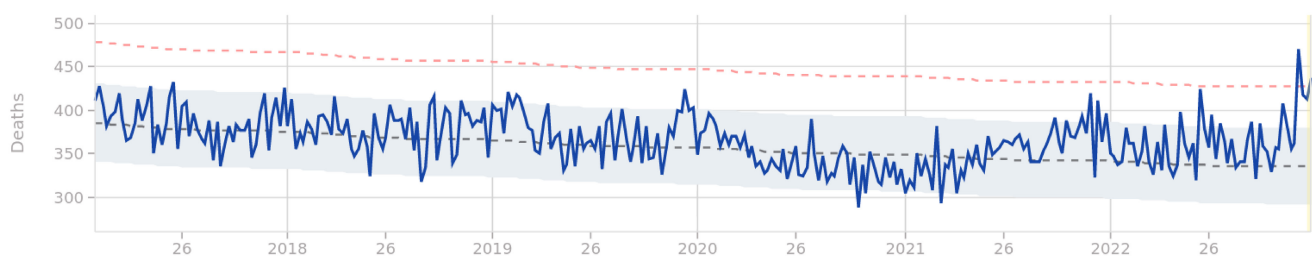


FIGURE 1 Graphs showing the weekly excess deaths (deviation in mortality from the expected level) (a) and the pooled weekly total number of deaths (b) from year 2017 to the end of 2022, in the age group 0–14, generated with data from 27 participating European countries (EuroMOMO 2022). [Correction added on 20 February 2023, after first online publication: Figure 1 and its caption have been changed to show the weekly excess deaths from 2017 and added the pooled weekly total number of deaths from 2017 to 2020.]

COVID-19 hospitalization prevented. According to reports, from January 2021 to the time of writing, 697 professional or semi-professional athletes, still active in their sport, under the age of 50, collapsed and died due to cardiovascular complications, such as primary cardiac arrest or electromechanical dissociation from suspected venous thromboembolism or other cause.⁸ We stratified by age and found 558 athletes were ≤ 35 years and 139 were over 35 years. Such number of documented events is likely largely underestimated, because of the difficulty to gather data from countries where the script/alphabet is not translatable. Notably, in a 38-years timespan (1966–2004), 1101 athletes under the age of 35 died ($\sim 29/\text{year}$) due to various heart-related conditions, 50% of whom had congenital anatomical heart disease and cardiomyopathies and 10% had atherosclerotic heart disease with early onset.⁹ The finding of 697 deaths from 2021 forward is concerning for a secular change and these data suggest a research registry for athletes with sudden unexpected cardiac arrest should be initiated with particular attention to COVID-19 vaccine administration data. [Correction added on 20 February 2023, after first online publication: The preceding sentences were updated to report the deaths of athletes aged

under 50 years old, caused by cardiovascular complications.] According to a study done on 301 teenagers between the ages of 13 and 18 who had received two doses of the Pfizer/BioNTech vaccine, 29.24% of participants experienced cardiovascular complications such as tachycardia, palpitations and 2.33% suffered myopericarditis.¹⁰ It is noteworthy, that no statistically significant increase in the incidence of myocarditis or pericarditis was observed in un-vaccinated subjects after SARS-CoV-2 infection, in a large population study.¹¹ According to the EuroMOMO data from 27 participating countries, there has been a substantial increase in young age excess mortality since the end of 2021 as shown for the example of the 0–14 age group in Figure 1 (although we note that the linear assumptions used by EuroMOMO to calculate excess mortality may introduce some bias in cumulating data).¹² [Correction added on 20 February 2023, after first online publication: The preceding sentence was updated to indicate that the linear assumptions were taken from EuroMOMO data.]

Furthermore, a German study looked at the specific subtype of IgG antibodies that the participants had produced over time, after receiving multiple mRNA vaccinations.¹³ Months after the second dose, the anti-spike antibodies

were increasingly composed of non-inflammatory IgG4, which substantially increased after the third dose and/or SARS-CoV-2 infection. Comparatively to the other three antibodies of its class, IgG4 plays a limited role in the immune process against viral or bacterial invaders.¹⁴ IgG4 has been found to become more common in people who are chronically exposed to allergens, because it elicits a lower immune response. This suggests that multiple mRNA vaccinations induce an “immune tolerance” towards SARS-CoV-2, even in age groups and naturally immune who are perfectly capable of dealing with the virus, in the vast majority of cases. [Correction added on 20 February 2023, after first online publication: the preceding paragraph was added.]

In conclusion we thank our colleagues for advancing the discourse on the extremely concerning safety data after COVID-19 vaccination, which prompt us to emphasize again and more firmly the need of biodistribution studies as well as of rational harm-benefit assessments by age group.

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CONFLICT OF INTEREST

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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